

PRD-001 — Authentication

Product: PrepHub V2

Status: Approved for implementation

Primary audience: Product, design, engineering, and QA

Depends on: PRD-000 — PrepHub Design System

Related documents:

- PRD-002 — District Verification
 - PRD-003 — Billing & Subscriptions
 - PRD-009 — Profile and Account Settings
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1. Purpose

This document defines how students create, access, recover, and delete their PrepHub accounts.

Authentication establishes the identity of the student. It must remain separate from the method through which the student receives access to PrepHub.

A student may receive PrepHub access through:

- A partnered school or district
- An individual paid subscription

Changing the student's access method must never create a new account or separate the student from their existing progress.

2. Core Architecture Principle

PrepHub must treat the following as separate concepts:

Account Identity

Who the student is and how they sign in.

Examples:

- Email and password
- Google Sign-In

Platform Access

Why the student is permitted to use PrepHub.

Examples:

- District-sponsored access
- Individual subscription

Student Learning Data

The student's permanent PrepHub history.

Examples:

- Diagnostic results
- Practice history
- Skill progress
- Study time
- Activity statistics
- Saved preferences

A student's learning data must remain attached to the same account even if the student changes their login method or access method.

3. Goals

The authentication system must:

- Allow students to create an account using email and password.
- Allow students to create an account using Google.
- Allow multiple login methods to connect to the same account.
- Keep students signed in after they close the browser.
- Support simultaneous sessions across multiple devices.
- Provide secure password recovery.
- Collect only the personal information required for the product.
- Collect the student's first name and grade level.
- Support self-service account deletion.
- Route new students into the access-selection flow.
- Remain simple enough for a high school student to complete without assistance.

4. Non-Goals

PRD-001 does not define:

- School or district search
- School-email verification
- District-domain matching
- Pricing plans
- Checkout
- Payment processing
- Subscription management
- Parent accounts
- Teacher accounts
- District administrator accounts
- Usernames
- Custom profile-photo uploads
- Accounts for users under age 13

These features are addressed in separate product documents.

5. Target Users

PrepHub V2 is designed for students in:

- 9th grade
- 10th grade
- 11th grade
- 12th grade

Users must confirm that they are at least 13 years old.

PrepHub V2 will not support accounts for children under age 13 during Version 1.

6. Locked Product Decisions

The following decisions are final for Version 1:

- Students may sign up using email and password.

- Students may sign up using Google.
 - First-time Google Sign-In automatically creates a PrepHub account.
 - Google and email/password authentication may be connected to the same account when they use the same verified email address.
 - A student who originally joined through Google may later create a password.
 - Password recovery occurs through an emailed reset link.
 - Students remain logged in after closing and reopening the browser.
 - Students may remain logged in on multiple devices simultaneously.
 - Login email and verified school email are stored separately.
 - PrepHub does not use usernames.
 - PrepHub asks only for the student's first name.
 - PrepHub does not require or request the student's last name during signup.
 - Students may edit their first name later.
 - Students may delete their own account.
 - PrepHub uses the student's Google profile photo when available.
 - Students without a Google photo are represented by their first initial.
 - Custom profile-photo uploads are not included in Version 1.
 - Grade level is collected during signup.
 - Grade options are limited to grades 9 through 12.
 - Authentication, district eligibility, and billing are independent systems attached to the same student account.
 - New students continue to the access-selection page after account creation.
 - The Privacy Policy and Terms of Service are linked during signup.
 - Students must confirm that they are at least 13 years old.
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7. Authentication Methods

7.1 Email and Password

Students may create and access a PrepHub account using:

- Login email
- Password

Passwords must never be stored as readable text.

PrepHub must use a trusted authentication provider or a modern password-hashing method.

7.2 Google Sign-In

Students may select:

Continue with Google

Google authentication may provide:

- Verified email address
- Display name
- Given name or first name
- Profile photo
- Google provider account identifier

PrepHub should use the student's Google-provided first name when available.

If Google only provides a complete display name, PrepHub may use the first portion as an initial first-name value, but the student must be allowed to edit it during account completion.

PrepHub must not require a last name.

7.3 Connected Login Methods

A single PrepHub account may support both:

- Email/password login
- Google login

Example:

A student initially creates an account using:

`student@example.com`

If the student later selects Google Sign-In using the same verified email address, Google should be connected to the existing PrepHub account instead of creating a duplicate account.

The student's existing information must remain unchanged, including:

- Diagnostic results
- Practice progress
- Activity history
- District access
- Subscription status

- Study statistics
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8. Email and Password Signup Page

8.1 Page Purpose

The signup page allows a new student to create a PrepHub account.

The page must feel:

- Modern
- Minimal
- Trustworthy
- Student-friendly
- Visually clean

The page must follow the desktop-first and responsive requirements defined in PRD-000.

8.2 Page Content

Title

Create your PrepHub account

Required Fields

- First Name
- Email
- Password
- Grade

Do not include:

- Last Name
- Username
- School
- District
- School Email

- Phone Number
- Address

Grade Options

- 9th Grade
- 10th Grade
- 11th Grade
- 12th Grade

No grade should be selected by default.

Primary Button

Create Account

Authentication Alternative

Display a visual divider followed by:

Continue with Google

Existing Account Link

Display:

Already have an account? Log in

8.3 Legal Acknowledgment

The signup page must display:

By creating an account, you agree to PrepHub's Terms of Service and acknowledge the Privacy Policy.

"Terms of Service" and "Privacy Policy" must each be clickable.

The student must also actively select:

I confirm that I am at least 13 years old.

This confirmation must not be selected by default.

The account must not be created until the student confirms that they are at least 13 years old.

PrepHub must record the versions of the Terms of Service and Privacy Policy acknowledged at account creation.

9. Email and Password Signup Flow

9.1 Standard Flow

1. The student opens the signup page.
2. The student enters their first name.
3. The student enters their email address.
4. The student creates a password.
5. The student selects their grade.
6. The student confirms that they are at least 13 years old.
7. The student selects **Create Account**.
8. PrepHub validates all required information.
9. PrepHub creates the student account.
10. PrepHub creates a persistent authenticated session.
11. PrepHub routes the student to the access-selection screen.

The access-selection screen is defined in PRD-002.

9.2 Personal Email Verification

Version 1 does not require a separate personal-email verification screen before the student enters the access-selection flow.

School-email verification is separate and is defined in PRD-002.

PrepHub may send account, security, and password-recovery messages to the student's login email.

10. Google Signup Flow

10.1 Standard Flow

1. The student selects **Continue with Google**.
 2. PrepHub begins the Google authentication process.
 3. The student authenticates through Google.
 4. Google returns a verified account identity.
 5. PrepHub checks whether the verified Google email is already associated with a PrepHub account.
 6. If no account exists, PrepHub automatically creates a provisional account.
 7. PrepHub imports the student's first name and profile photo when available.
 8. PrepHub displays the account-completion screen.
 9. The student reviews or edits their first name.
 10. The student selects their grade.
 11. The student confirms that they are at least 13 years old.
 12. The student acknowledges the Terms of Service and Privacy Policy.
 13. PrepHub completes account creation.
 14. PrepHub creates a persistent authenticated session.
 15. PrepHub routes the student to the access-selection screen.
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10.2 Account-Completion Screen

Google does not provide the student's grade or age confirmation. Therefore, first-time Google users must complete one minimal follow-up screen.

Title

Finish setting up your account

Fields and Information

- First Name
- Google Email
- Grade
- Age confirmation
- Terms of Service and Privacy Policy acknowledgment

The first name should be prefilled when Google provides one.

The student may edit the first name before continuing.

The Google email should be visible but not editable on this screen.

Do not ask for:

- Last name
- Username
- School
- School email
- Phone number
- Additional demographic information

Primary Button

Continue

10.3 Existing Email Account

If a student already has an email/password account using the same verified Google email:

- Do not create another student account.
- Connect the Google identity to the existing account.
- Preserve all existing student data.
- Sign the student into the existing account.

The system must prevent duplicate accounts caused by using different login methods with the same verified email.

11. Login Page

11.1 Page Content

Title

Welcome back

Fields

- Email
- Password

Primary Button

Log In

Additional Actions

- Continue with Google
 - Forgot Password?
 - Don't have an account? Create one
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11.2 Successful Login Routing

After login, PrepHub must determine the appropriate destination.

Active Access

If the student has active district-sponsored access or an active individual subscription:

- Route the student to the Student Dashboard.

No Active Access

If the student has an account but has not selected or completed an access method:

- Route the student to the access-selection page.

Pending District Verification

If the student has started but not completed school verification:

- Route according to the rules in PRD-002.

Incomplete Individual Purchase

If the student selected individual access but does not have an active subscription:

- Route according to PRD-003.

Authentication must not independently grant district-sponsored or paid access.

12. Forgot Password

12.1 Entry Point

The login page must include:

Forgot Password?

12.2 Flow

1. The student selects **Forgot Password?**
 2. PrepHub displays an email input.
 3. The student enters their login email.
 4. The student selects **Send Reset Link**.
 5. PrepHub sends a secure password-reset link when an eligible account exists.
 6. The student opens the reset link.
 7. The student creates and confirms a new password.
 8. PrepHub invalidates the reset token after use.
 9. The student may log in using the new password.
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12.3 Confirmation Message

To reduce account-discovery risks, the confirmation should use neutral wording:

If an account exists for that email, we've sent a password-reset link.

The screen must not reveal whether an email is registered.

12.4 Google-Created Accounts

A student who originally created their account through Google may use the password-reset process to create a password for the same email address.

After creating a password, the student may sign in using either:

- Google

- Email and password

Creating a password must not create another PrepHub account.

13. Session Management

13.1 Persistent Login

Students should remain logged in after:

- Closing the browser
- Reopening the browser
- Restarting their device

The system must use secure persistent authentication.

PrepHub must never store the student's readable password in the browser.

13.2 Multiple Devices

Students may remain signed in simultaneously on multiple devices, including:

- School Chromebook
- Personal laptop
- Desktop computer
- Tablet
- Phone

Signing in on a new device must not automatically end existing sessions.

13.3 Logout

Selecting **Log Out** ends the PrepHub session on the current device.

Other device sessions remain active unless:

- The account is deleted

- The account is disabled
 - Sessions are revoked for security reasons
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13.4 Session Security

Sessions must:

- Use secure and unpredictable tokens.
 - Be revocable.
 - Expire according to a configurable security policy.
 - Be invalidated after account deletion.
 - Avoid storing sensitive authentication information in readable browser storage.
 - Use secure browser-cookie settings when cookies are used.
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14. Student Identity Data

14.1 Required Account Information

Each student account must contain:

- First name
- Primary login email
- Grade level
- Account creation date
- Account status

PrepHub must not require a last name for core account functionality.

14.2 Editable Information

Students may later edit:

- First name
- Grade

The editing interface will be defined in PRD-009.

Changing either field must not create a new account or affect the student's progress.

14.3 Usernames

PrepHub will not create or display usernames in Version 1.

Students are identified internally through a permanent, system-generated user ID.

15. Profile Image Rules

15.1 Google Profile Photo

When available, PrepHub may display the profile photo provided by Google.

The photo is optional and must not be required for account creation.

15.2 First-Initial Fallback

When no Google photo is available, PrepHub should display the first letter of the student's first name.

Example:

Prithviraj → **P**

The fallback avatar should be generated automatically.

15.3 Custom Uploads

Students cannot upload custom profile photos in Version 1.

This avoids unnecessary:

- Content moderation

- File storage
 - Image-security concerns
 - Student-privacy concerns
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16. Login Email and School Email

PrepHub must store login identity and school eligibility separately.

Example:

Login Email

`student@gmail.com`

Used for:

- Signing in
- Password recovery
- Account communication

Verified School Email

`student@k12.friscoisd.org`

Used for:

- Proving district eligibility
- Linking district-sponsored access

The verified school email must not replace the login email unless the student originally used that school email to create their account.

A school email may be linked to only one PrepHub account.

Detailed school-email rules are defined in PRD-002.

17. Account Deletion

17.1 Availability

Students must be able to initiate account deletion without contacting support.

The deletion control will appear in Profile or Account Settings.

17.2 Confirmation

Account deletion must require clear confirmation.

Email/Password Accounts

Require password reauthentication before deletion.

Google-Only Accounts

Require Google reauthentication before deletion.

The confirmation screen must clearly explain that deleting the account affects:

- Practice history
 - Diagnostic results
 - Skill progress
 - Activity statistics
 - Saved study history
 - District association
 - Subscription association
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17.3 Deletion Behavior

After confirmed deletion:

- Revoke all active sessions.
- Disable access to the account.
- Remove or anonymize personal information according to PrepHub's legal and retention requirements.
- Prevent continued district-sponsored or paid access through the deleted account.
- Follow the policies described in the Privacy Policy.

The exact retention period, if one is required, will be defined in PrepHub's legal and data-retention documentation.

18. Validation Requirements

18.1 First Name

The first-name field must:

- Be required.
- Remove leading and trailing spaces.
- Support legitimate names containing spaces, hyphens, apostrophes, and accented characters.
- Not assume every first name follows English naming conventions.
- Not require a minimum longer than necessary to support legitimate short names.

The label should be:

First Name

PrepHub may use the entered first name for:

- Dashboard greetings
 - Support identification
 - First-initial avatar generation
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18.2 Email

The email field must:

- Be required.
 - Use valid email formatting.
 - Be normalized for account comparison.
 - Be compared without depending on letter capitalization.
 - Not create a second account when an equivalent email is already registered.
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18.3 Password

The password must:

- Be required for email/password signup.
 - Meet reasonable security requirements.
 - Have its requirements displayed clearly.
 - Never appear in application logs or product analytics.
 - Be processed through a trusted authentication provider or secure hashing system.
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18.4 Grade

Grade is required and must be one of:

- 9
 - 10
 - 11
 - 12
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18.5 Age Confirmation

The age confirmation must:

- Be required.
 - Not be selected by default.
 - Prevent account completion when it is not selected.
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19. Error States

All errors must use plain, friendly language.

19.1 Email Already Registered

Message:

An account already exists with this email. Try logging in instead.

Provide a direct link to Login.

19.2 Incorrect Login Information

Message:

The email or password you entered is incorrect.

Do not reveal which field was incorrect.

19.3 Google Authentication Cancelled

Message:

Google Sign-In was cancelled. You can try again or use email and password.

19.4 Google Authentication Failure

Message:

We couldn't sign you in with Google. Please try again.

19.5 Network Failure

Message:

We couldn't connect to PrepHub. Check your connection and try again.

19.6 Expired Password-Reset Link

Message:

This password-reset link has expired. Request a new one to continue.

Include a button:

Request New Link

19.7 Incomplete Required Information

Display validation near the relevant field.

Do not erase valid information the student has already entered.

20. Loading and Interaction States

Buttons must show a loading state after submission.

Examples:

- Creating Account...
- Signing In...
- Sending Link...
- Resetting Password...

During submission:

- Prevent accidental duplicate requests.
- Keep the page responsive.
- Preserve valid information when recoverable.
- Do not show repeated or unnecessary popups.

Animations must remain subtle and professional.

21. Logical Data Requirements

The exact physical database implementation may vary, particularly when PrepHub uses a managed authentication provider.

The system must support the following logical records.

21.1 User Account

Required fields:

- `id`
- `first_name`
- `primary_login_email`
- `grade_level`
- `profile_photo_url`
- `account_status`
- `created_at`
- `updated_at`
- `deleted_at`

Do not require a `last_name` field.

The `id` must be the permanent internal identifier connecting all PrepHub student data.

21.2 Authentication Identity

Required fields:

- `id`
- `user_id`
- `provider`
- `provider_user_id`
- `provider_email`
- `provider_email_verified`
- `created_at`
- `last_used_at`

Supported provider values for Version 1:

- `password`
- `google`

One user may have multiple authentication identities.

21.3 Password Credential

When not entirely managed by an external authentication provider, the logical record must support:

- `user_id`
- `password_hash`
- `password_updated_at`

Readable passwords must never be stored.

21.4 Session

Required fields:

- `id`
 - `user_id`
 - Secure token hash or provider equivalent
 - `created_at`
 - `expires_at`
 - `last_active_at`
 - `revoked_at`
 - Reasonable device or browser metadata when required for security
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21.5 Password Reset

Required fields:

- `user_id`
- `token_hash`
- `created_at`
- `expires_at`
- `used_at`

Password-reset tokens must:

- Expire
- Be single-use

- Be securely generated
 - Not be stored as readable tokens
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21.6 Legal Acceptance

Required fields:

- `user_id`
- `terms_version`
- `privacy_policy_version`
- `accepted_at`
- `age_13_or_older_confirmed`
- Reasonable technical metadata required for the acceptance record

PrepHub must be able to identify which document versions the student acknowledged.

22. Separation From Access and Billing

The user-account record must not use a single permanent field such as:

- `is_paid`
- `is_district_student`
- `has_access`

as the sole source of access truth.

Instead, access must be determined through separate entitlement records.

Examples:

- District entitlement
- Individual subscription entitlement
- Administrative or promotional entitlement, if introduced later

This allows a student to:

- Begin with an individual subscription.
- Later verify district-sponsored access.
- End the individual subscription.

- Continue using the same PrepHub account.
- Preserve all learning history throughout the transition.

Authentication proves identity. It does not independently determine platform access.

23. Security Requirements

The authentication system must:

- Never store readable passwords.
 - Use a trusted authentication library or provider.
 - Use secure password hashing.
 - Use HTTPS in production.
 - Protect Google OAuth requests with appropriate security controls.
 - Use Google's provider identifier rather than email alone as the permanent Google identity.
 - Rate-limit repeated login attempts.
 - Rate-limit repeated password-reset attempts.
 - Prevent duplicate accounts using equivalent email addresses.
 - Protect session tokens.
 - Revoke sessions after account deletion.
 - Avoid exposing whether an account exists during password recovery.
 - Prevent secrets from appearing in logs, analytics, or error messages.
 - Record security-relevant events when appropriate.
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24. Privacy Requirements

PrepHub should collect only information necessary for:

- Account creation
- Authentication
- Personalization
- Student progress
- District eligibility
- Subscription access
- Product security
- Necessary operational analytics

PrepHub should not collect a last name merely because it may be useful later.

Authentication data must not be sold to advertisers.

The Privacy Policy must explain:

- What information PrepHub collects
- Why the information is collected
- How the information is used
- Which service providers process it
- How account deletion works
- How students and families can contact PrepHub

District-specific student-data obligations will also be addressed through PRD-002 and district agreements.

25. Accessibility Requirements

All authentication screens must:

- Be fully usable with a keyboard.
 - Use properly associated input labels.
 - Provide visible focus states.
 - Support password-manager autofill.
 - Support appropriate browser autocomplete.
 - Maintain sufficient text and control contrast.
 - Announce validation errors to assistive technologies.
 - Avoid relying only on color to communicate errors.
 - Use readable text sizes.
 - Provide appropriately sized mobile touch targets.
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26. Responsive Design Requirements

PrepHub is desktop-first and fully responsive.

Desktop

- Center the authentication form in a clean page layout.

- Keep input widths comfortable.
- Avoid stretching the form across the entire screen.
- Maintain generous but practical spacing.

Tablet

- Preserve the desktop content hierarchy.
- Reduce surrounding whitespace where appropriate.

Mobile

- Use a single-column layout.
- Keep fields and buttons full-width where appropriate.
- Avoid horizontal scrolling.
- Ensure the on-screen keyboard does not hide the active field or primary action.

No authentication function may be unavailable on mobile.

27. Analytics Events

PrepHub may record product analytics for:

- Signup page viewed
- Email signup started
- Email signup completed
- Google signup started
- Google signup completed
- Login completed
- Login failed
- Password reset requested
- Password reset completed
- Account completion completed
- Account deletion initiated
- Account deletion completed

Analytics must never include:

- Passwords
- Password-reset tokens
- Session tokens

- OAuth credentials
 - Sensitive authentication content
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28. Acceptance Criteria

PRD-001 is complete when all the following requirements are satisfied.

Signup

- A student can create an account using email and password.
- Required fields are first name, email, password, and grade.
- No last-name field is displayed.
- Grade is limited to grades 9 through 12.
- The student must confirm that they are at least 13.
- Terms of Service and Privacy Policy links are visible.
- Successful signup creates an authenticated session.
- The student is routed to the access-selection page.

Google Authentication

- A student can create an account through Google.
- A first-time Google user receives a PrepHub account automatically.
- Google first name and profile photo are imported when available.
- The student may review or edit the imported first name.
- No last name is required.
- The student completes grade and legal acknowledgment before continuing.
- Google Sign-In connects to an existing account when the verified email matches.
- Duplicate student accounts are not created.

Login

- Students can log in using email and password.
- Students can log in using Google.
- Active users are routed according to their access state.
- Students remain logged in after closing the browser.
- Students can remain logged in on multiple devices.

Password Recovery

- Students can request a password-reset email.
- Reset links expire and are single-use.
- Google-created users can create a password through the reset flow.
- Creating a password does not create a duplicate account.

Identity

- Login email and verified school email can be stored separately.
- No usernames are created.
- No last name is required.
- Google photo is used when available.
- The student's first initial is shown otherwise.
- Custom profile-photo uploads are not available.

Account Management

- Students can later edit their first name and grade.
- Students can initiate account deletion.
- Account deletion requires reauthentication.
- Account deletion revokes active sessions.
- Student progress remains attached to the permanent account regardless of access method.

Quality

- All screens follow PRD-000.
 - All screens work on desktop, tablet, and mobile.
 - Validation messages are clear.
 - Loading and error states are implemented.
 - Authentication secrets are never exposed or stored insecurely.
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29. Future Considerations

The following are intentionally excluded from Version 1:

- Parent or guardian accounts
- Teacher authentication
- District administrator single sign-on
- Microsoft Sign-In

- School-managed SSO
- Passkeys
- Two-factor authentication
- Custom student profile photos
- Student last names unless a future district requirement creates a clear need
- User-facing device and session management
- Accounts for children under 13 with verified parental consent

These features must not be added during Version 1 implementation without a separate product decision.